

# Engineering Coding Test: Habit Shaper

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## Overview

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Build a **Habit Shaper** — a lightweight, web-based application that helps users build positive habits and break negative ones through daily tracking and streak management. This test evaluates your ability to plan, implement, and deliver a complete end-to-end system from a set of requirements.

## Requirements

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### Core Features

#### 1. Habit Building

- Users can create habits they want to build (e.g., meditate, exercise, read, learn Mandarin)
- Users can mark habits as completed for each day
- Track streaks — consecutive days of completion
- Track weekly completion rate — show how many days were missed per week

#### 2. Habit Breaking

- Users can create habits they want to break (e.g., smoking, junk food, doomscrolling)
- Track "clean" streaks — consecutive days without the habit (day 1, day 2, day 3, ...)
- Reset streak when the user relapses

#### 3. Goal Management

- Users can add, edit, and remove goals
- Each goal is linked to a habit — either a "build" or "break" type

### Technical Specification

- **Web-based** application
- **Lightweight** — keep the theme and design simple and clean
- **Tech stack:**
  - **Frontend:** React
  - **Backend:** Node.js with TypeScript

- **Database:** MySQL
- **Authentication:** Register and login via email and password (no email verification required)
- **Containerized setup** — the whole stack **must** be runnable with a single `docker compose up` from the repository root. No local runtime or database installation should be required.

## Submission

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- Create a **private** repository and invite the reviewer (provided in the email) as a collaborator
- Commit **everything** — source code, documentation, configurations, and any artifacts generated during your development process. We want to see how you arrived at the final result, not just the result itself.
- Include a working `compose.yml` (the current Docker Compose filename convention) at the repository root that starts the entire application. Schema bootstrap (migrations or seed files) must run automatically on first boot.
- Exclude `node_modules`, build artifacts, and any real secrets ( `.env`, certs, keys). Provide a `.env.example` with placeholder values and document any required variables in `README.md`.
- The app should be runnable with only **Docker + Docker Compose** installed — no additional toolchain setup. Include the exact commands in `README.md`.

## Evaluation Criteria

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| Aspect                     | What We Look For                                                    |
|----------------------------|---------------------------------------------------------------------|
| <b>Agentic Development</b> | Effective use of coding agents (Claude Code, Copilot, Cursor, etc.) |
| <b>Code Quality</b>        | Clean, readable, well-structured code                               |
| <b>Completeness</b>        | All core features working end-to-end                                |
| <b>Git History</b>         | Meaningful commits that tell the story of how you built it          |